

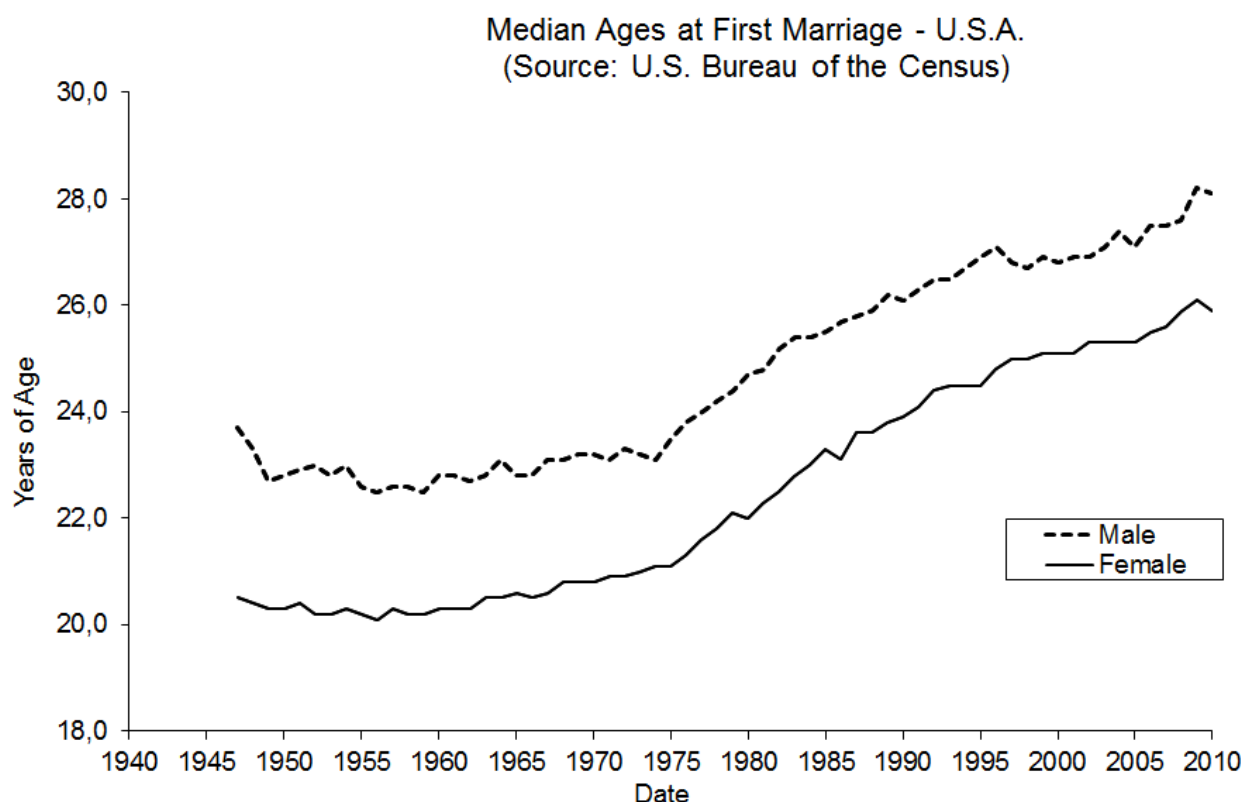
**Advanced Econometrics – Exam**

September 3, 2018

**Exercise 1**

Dave Giles at May 30, 2011 published a dataset with estimated median age of Females and Males at first marriage in the USA. In the dataset median age for each year from 1947 to 2010 is stored. The blogger suggested that possible cointegration between the median age for Females (Female) and Males (Male) may exist. To assess that, a number of analyses were performed. Furthermore, it was decided to check whether an Error Correction Mechanism model is appropriate here. Based on the results presented below please answer the following questions:

- Looking at the figure presented below, is there any rationality to consider a possibility that the Female and Male variables are cointegrated?
- Do time series of Female and Male variables satisfy basic assumptions to consider a cointegration between them (please support your answer with appropriate tests results)?
- Is there enough evidence to support a claim that there is cointegration between the variables (please support your answer with appropriate tests results)?
- What is or would be a cointegrating vector for the variables? How to interpret it?
- Does ECM model work? How would you interpret coefficients in the ECM model?



| Female median age at first marriage |      |       |                |                |                |                               |
|-------------------------------------|------|-------|----------------|----------------|----------------|-------------------------------|
| augmentations                       | adf  | p_adf | p_bg.p.value.1 | p_bg.p.value.2 | p_bg.p.value.3 | p_bg.p.value.4 p_bg.p.value.5 |
| 0                                   | 1.84 | 0.99  | 0.25           | 0.19           | 0.22           | 0.10 0.11                     |
| 1                                   | 1.95 | 0.99  | 0.99           | 0.33           | 0.30           | 0.17 0.18                     |
| 2                                   | 1.29 | 0.99  | 0.77           | 0.94           | 0.80           | 0.70 0.44                     |
| 3                                   | 0.94 | 0.99  | 0.79           | 0.84           | 0.89           | 0.63 0.49                     |
| 4                                   | 0.49 | 0.98  | 0.83           | 0.63           | 0.68           | 0.81 0.47                     |
| 5                                   | 0.20 | 0.97  | 0.55           | 0.76           | 0.86           | 0.90 0.87                     |

| First differences for female median age at first marriage |       |       |                |                |                |                               |
|-----------------------------------------------------------|-------|-------|----------------|----------------|----------------|-------------------------------|
| augmentations                                             | adf   | p_adf | p_bg.p.value.1 | p_bg.p.value.2 | p_bg.p.value.3 | p_bg.p.value.4 p_bg.p.value.5 |
| 0                                                         | -8.45 | 0.01  | 0.92           | 0.16           | 0.13           | 0.07 0.09                     |
| 1                                                         | -4.67 | 0.01  | 0.66           | 0.84           | 0.73           | 0.68 0.40                     |
| 2                                                         | -3.42 | 0.02  | 0.72           | 0.72           | 0.73           | 0.56 0.46                     |
| 3                                                         | -2.29 | 0.21  | 0.79           | 0.56           | 0.57           | 0.69 0.43                     |
| 4                                                         | -2.19 | 0.25  | 0.54           | 0.74           | 0.84           | 0.87 0.86                     |
| 5                                                         | -1.88 | 0.36  | 0.79           | 0.90           | 0.97           | 0.99 0.99                     |

| Male median age at first marriage |      |       |                |                |                |                               |
|-----------------------------------|------|-------|----------------|----------------|----------------|-------------------------------|
| augmentations                     | adf  | p_adf | p_bg.p.value.1 | p_bg.p.value.2 | p_bg.p.value.3 | p_bg.p.value.4 p_bg.p.value.5 |
| 0                                 | 1.33 | 0.99  | 0.86           | 0.88           | 0.28           | 0.43 0.56                     |
| 1                                 | 1.21 | 0.99  | 0.33           | 0.55           | 0.15           | 0.21 0.32                     |
| 2                                 | 1.32 | 0.99  | 0.94           | 0.96           | 0.06           | 0.11 0.18                     |
| 3                                 | 0.82 | 0.99  | 0.92           | 0.94           | 0.89           | 0.95 0.98                     |
| 4                                 | 0.67 | 0.99  | 0.86           | 0.98           | 0.98           | 0.98 1.00                     |
| 5                                 | 0.56 | 0.99  | 0.81           | 0.96           | 0.99           | 0.97 0.99                     |

| First differences for male median age at first marriage |       |       |                |                |                |                               |
|---------------------------------------------------------|-------|-------|----------------|----------------|----------------|-------------------------------|
| augmentations                                           | adf   | p_adf | p_bg.p.value.1 | p_bg.p.value.2 | p_bg.p.value.3 | p_bg.p.value.4 p_bg.p.value.5 |
| 0                                                       | -7.90 | 0.01  | 0.33           | 0.59           | 0.11           | 0.14 0.23                     |
| 1                                                       | -6.82 | 0.01  | 0.89           | 0.98           | 0.04           | 0.07 0.13                     |
| 2                                                       | -4.08 | 0.01  | 0.86           | 0.95           | 0.91           | 0.97 0.99                     |
| 3                                                       | -3.00 | 0.04  | 0.84           | 0.98           | 0.99           | 0.98 0.99                     |
| 4                                                       | -2.09 | 0.29  | 0.82           | 0.96           | 0.99           | 0.96 0.99                     |
| 5                                                       | -2.19 | 0.25  | 0.98           | 0.79           | 0.91           | 0.96 0.98                     |

## Results for the Long-term relationship model

```

=====
Dependent variable:
-----
Female
-----
Male                1.102***
                   (0.017)

Constant            -4.808***
                   (0.417)

-----
Observations                64
R2                          0.986
Adjusted R2                 0.986
Residual Std. Error    0.249 (df = 62)
F Statistic             4,291.233*** (df = 1; 62)
=====
Note:                *p<0.1; **p<0.05; ***p<0.01

```

## Residuals from Long-term relationship model

| augmentations | adf   | p_adf | p_bg.p.value.1 | p_bg.p.value.2 | p_bg.p.value.3 | p_bg.p.value.4 | p_bg.p.value.5 |
|---------------|-------|-------|----------------|----------------|----------------|----------------|----------------|
| 0             | -5.38 | 0.01  | 0.57           | 0.72           | 0.42           | 0.59           | 0.73           |
| 1             | -4.28 | 0.01  | 0.95           | 0.97           | 0.70           | 0.79           | 0.88           |
| 2             | -2.82 | 0.07  | 0.94           | 0.99           | 0.95           | 0.91           | 0.95           |
| 3             | -2.56 | 0.11  | 0.96           | 1.00           | 1.00           | 0.99           | 0.99           |
| 4             | -2.11 | 0.28  | 0.94           | 0.93           | 0.98           | 0.99           | 0.99           |
| 5             | -2.27 | 0.22  | 0.80           | 0.97           | 0.99           | 0.99           | 0.98           |

## Results for the ECM model

```

=====
Dependent variable:
-----
dFemale
-----
lresid              -0.270***
                   (0.070)

dMale               0.307***
                   (0.082)

Constant            0.068***
                   (0.017)

-----
Observations                63
R2                          0.255
Adjusted R2                 0.230
Residual Std. Error    0.125 (df = 60)
F Statistic             10.280*** (df = 2; 60)
=====
Note:                *p<0.1; **p<0.05; ***p<0.01

```

**Exercise 3 (40%)**

Dear Students, please meet mink and muskrat.

Figure 1. Muskrat



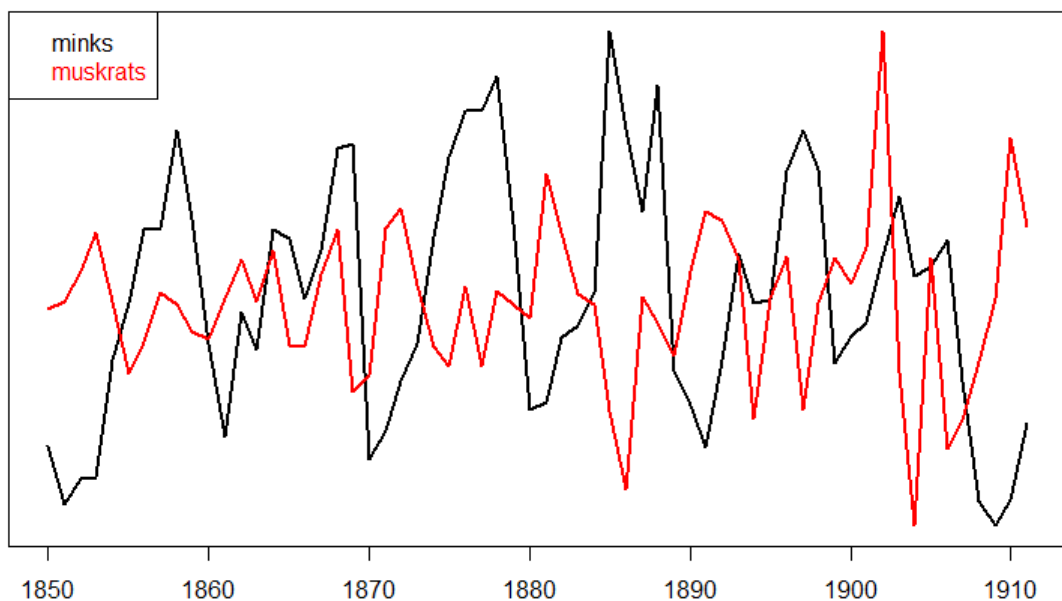
Source: [wikimedia.org/ \(link\)](#)

Figure 2. Mink



Source: [wikimedia.org/ \(link\)](#)

**Mink** are dark-colored, semiaquatic, carnivorous mammals of the genera *Neovison* and *Mustela*, and part of the family *Mustelidae*, which also includes weasels, otters and ferrets. The American mink's fur has been highly prized for use in clothing, with hunting giving way to farming.<sup>5</sup> The **muskrat** (*Ondatra zibethicus*), the only species in genus *Ondatra* and tribe *ondatrini*, is a medium-sized semiaquatic rodent native to North America and an introduced species in parts of Europe, Asia, and South America. Muskrats provide an important food resource for many other animals, including **mink**, foxes, coyotes, wolves, lynx, bobcats, bears, eagles, snakes, alligators, and large owls and hawks.<sup>6</sup>



[Let us now discuss] the two series shown in Figure 3. These series record the numbers of mink

<sup>5</sup><https://en.wikipedia.org/wiki/Mink>

<sup>6</sup><https://en.wikipedia.org/wiki/Muskrat>

and muskrat furs traded annually by the Hudson's Bay Company in Canada from 1848 to 1909. There is known to be a prey-predator relationship between these two animals and this affects the population dynamics of both of them.<sup>7</sup>

Our goal will be to establish a relationship between number of mink and muskrats. The output below consists of R output obtained from various time series models and analyses.

- a) (10%) Let's now inspect the series visually. Are they integrated? Explain.
- b) (10%) What is the integration order of `minks` variables? Explain.
- c) (5%) Are `minks` and `muskrats` cointegrated? Explain.
- d) (5%) Which model should be selected out of the three ARDL models presented below? Explain.
- e) (10%) What does the Granger-causality analysis indicate? Explain.

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<sup>7</sup>A.C.Harvey, (1989), *Forecasting, Structural Time Series and the Kalman Filter*, Cambridge University Press, p. 9.

```

> testdf2(variable = dt$minks, test.type="nc",
+         max.augmentations = 4, max.order = 3)
  augmentations      adf      p_adf bgodfrey.statistic bgodfrey.statistic.1
1             0 -1.0434878 0.2800023             61             0.045580522
2             1 -1.0286117 0.2847290             60             0.005477665
3             2 -0.8920156 0.3281307             59             0.022220094
4             3 -0.8334178 0.3467494             58             0.019830329
5             4 -0.7283168 0.3801440             57             0.225382612
bgodfrey.statistic.2 bgodfrey.statistic.3 p_bg.p.value.1 p_bg.p.value.2 p_bg.p.value.3
1             1.4057657             1.5736972             0.8309401             0.4951558             0.6653684
2             1.3370138             1.4198942             0.9410014             0.5124732             0.7008781
3             0.1392349             0.3560613             0.8815031             0.9327506             0.9491619
4             0.1761922             0.2301956             0.8880120             0.9156729             0.9725735
5             0.7724574             1.2685787             0.6349689             0.6796151             0.7366075

> testdf2(variable = dt$muskrats, test.type="nc",
+         max.augmentations = 4, max.order = 3)
  augmentations      adf      p_adf bgodfrey.statistic bgodfrey.statistic.1
1             0 -6.323482  0.01             61             0.13383622
2             1 -6.219694  0.01             60             0.05401426
3             2 -4.081416  0.01             59             0.08147088
4             3 -5.258275  0.01             58             0.19525076
5             4 -4.903783  0.01             57             0.22875325
bgodfrey.statistic.2 bgodfrey.statistic.3 p_bg.p.value.1 p_bg.p.value.2 p_bg.p.value.3
1             4.9190242             6.0195304             0.7144872             0.08547665             0.1106639
2             0.9545065             2.0253250             0.8162198             0.62048536             0.5671671
3             0.4988606             0.7649031             0.7753139             0.77924459             0.8578420
4             0.9183355             1.2441547             0.6585819             0.63180924             0.7424348
5             1.9146262             2.0947157             0.6324498             0.38392307             0.5529826

> testdf2(variable = diff(dt$minks), test.type="nc",
+         max.augmentations = 4, max.order = 3)
  augmentations      adf      p_adf bgodfrey.statistic bgodfrey.statistic.1
1             0 -7.464321  0.01             60             0.002562517
2             1 -6.093589  0.01             59             0.003487338
3             2 -4.906297  0.01             58             0.005505509
4             3 -5.035137  0.01             57             0.196935050
5             4 -5.147511  0.01             56             0.412764193
bgodfrey.statistic.2 bgodfrey.statistic.3 p_bg.p.value.1 p_bg.p.value.2 p_bg.p.value.3
1             1.34064025             1.4332867             0.9596273             0.5115448             0.6977510
2             0.08212202             0.3520402             0.9529093             0.9597706             0.9499615
3             0.13588402             0.1852202             0.9408520             0.9343147             0.9799392
4             0.73381946             1.2342153             0.6572062             0.6928722             0.7448103
5             1.58478627             2.7552686             0.5205698             0.4527600             0.4309165

```

```
> testdf2(variable = diff(dt$muskrats), test.type="nc",
+         max.augmentations = 4, max.order = 3)
      augmentations      adf p_adf bgodfrey.statistic bgodfrey.statistic.1
1           0 -10.042191  0.01           60           0.98382055
2           1 -10.171713  0.01           59           0.01475214
3           2  -5.690695  0.01           58           0.01099714
4           3  -5.463414  0.01           57           0.01866789
5           4  -4.328418  0.01           56           0.01579248
      bgodfrey.statistic.2 bgodfrey.statistic.3 p_bg.p.value.1 p_bg.p.value.2 p_bg.p.value.3
1          12.9114283      13.3956031      0.3212574      0.001571517      0.003854706
2           0.6313132      0.6958729      0.9033280      0.729309844      0.874174246
3           0.5801917      0.7266606      0.9164812      0.748191857      0.866915912
4           0.1421436      0.5131459      0.8913229      0.931395027      0.915993965
5           0.2775689      0.6409912      0.8999946      0.870415629      0.886987529

> model.coint = lm(minks~muskrats, data=dt)
> summary(model.coint)
```

Call:

```
lm(formula = minks ~ muskrats, data = dt)
```

Residuals:

| Min    | 1Q     | Median | 3Q    | Max   |
|--------|--------|--------|-------|-------|
| -34243 | -13125 | -1458  | 11450 | 33489 |

Coefficients:

|             | Estimate    | Std. Error | t value | Pr(> t )                |
|-------------|-------------|------------|---------|-------------------------|
| (Intercept) | 52091.16976 | 2211.27859 | 23.557  | <0.0000000000000002 *** |
| muskrats    | -0.02730    | 0.01069    | -2.554  | 0.0132 *                |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 17380 on 60 degrees of freedom

Multiple R-squared: 0.09809, Adjusted R-squared: 0.08306

F-statistic: 6.525 on 1 and 60 DF, p-value: 0.01319

```
> testdf2(variable = resid(model.coint), test.type="nc",
+         max.augmentations = 4, max.order = 3)
      augmentations      adf p_adf bgodfrey.statistic bgodfrey.statistic.1
1           0 -4.208015  0.01           61           0.243551850
2           1 -4.465550  0.01           60           0.014795433
3           2 -3.794224  0.01           59           0.009270986
4           3 -4.094188  0.01           58           0.072743384
5           4 -3.870033  0.01           57           0.086777440
      bgodfrey.statistic.2 bgodfrey.statistic.3 p_bg.p.value.1 p_bg.p.value.2 p_bg.p.value.3
1          1.4452028      1.4598256      0.6216525      0.4854877      0.6915731
2           0.2958177      0.4770259      0.9031869      0.8625097      0.9239089
3           0.1648981      0.2964598      0.9232935      0.9208584      0.9606929
4           0.2411588      0.2817348      0.7873835      0.8864067      0.9634263
5           0.2214667      0.2256841      0.7683147      0.8951774      0.9733402
```